From the fig. 8.3 we find that:

Total number of balls = 7+12+14=23 balls

Out of 23 balls 3 balls can be selected in = $n(s)= 23C_3$ ways

Let A be the event of choosing 3 balls which is blue

Number of possible ways of drawing 3 out of 12 blue balls is = $n(A)=12C_3$ ways

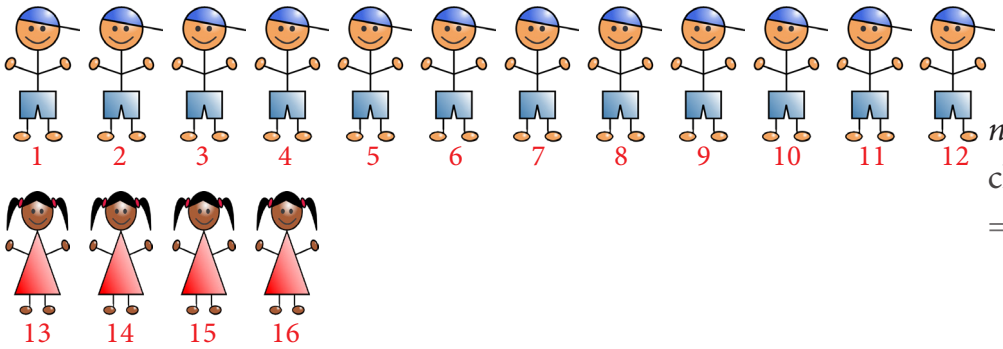
Therefore, $P(A) = \frac{n(A)}{n(S)} = \frac{12C_3}{23C_3} = \frac{220}{1771}$

$= 0.1242$

Example 8.7

A class has 12 boys and 4 girls. Suppose 3 students are selected at random from the class. Find the probability that all are boys.

Solution:



$n(s) = 16$

choosing 3 students

$= 16\ C_3$

chossing any 3 boys out of 12 boys is $12C_3 = 220$ ways

fig. 8.4 Selection of 3 Boys

From the fig 8.4, we find that:

Total number of students = 12+4=16

Three students can be selected out of 16 students in $16C_3$ ways

i.e. $n(s) = 16C_3 = \frac{16 \times 15 \times 14}{1 \times 2 \times 3} = 560$

Three boys can be selected from 12 boys in $12C_3$ ways

i.e. $n(A) = 12C3 = \frac{12 \times 11 \times 10}{1 \times 2 \times 3} = 220$

The required probability $P(A) = \frac{n(A)}{n(S)} = \frac{220}{560} = \frac{11}{28}$

$= 0.392$